

AEGIS Architecture

Anonymization & Exchange Gateway for Imaging Studies

SENDING SOURCES External Browser Upload · DICOM Network (C-STORE SCP, port 11112) · Programmatic Batch Ingest					
React Upload Portal (PWA) - Drag-and-drop: files, folders, DICOM directories - Client-side tag de-identification (DICOM PS3.15) - Anonymization preview — before / after tag diff - Multi-study detection with per-study upload cards - Per-project retained-tag anonymization profiles - Auto-retry: 3× exponential backoff per file		DIMSE Receiver (pynetdicom · GCE VM) - Compute Engine VM — aegis-prod-dimse-receiver - Static IP <GCP_DIMSE_PUBLIC_IP> · TCP port 11112 - C-STORE SCP — receives from PACS systems / scanners - Institution attribution (AE title or IP CIDR) - Calls POST /api/ingest on DICOM association close - Durable retry queue + dead-letter (disk-persistent)		Key Privacy Principle PHI is stripped in the browser BEFORE data leaves the hospital. - Only tag-de-identified DICOM is transmitted - HIPAA Safe Harbor — 18 identifier types removed - Deterministic UID hashing + date shifting - Zero software install required at sending site - DIMSE path ingests raw files (internal-only path)	
GCP PROJECT <GCP_PROJECT_ID> · us-central1 · Live, February 2026					
Cloud Armor (DDoS / WAF) + Global HTTPS Load Balancer + Identity-Aware Proxy (admin routes) + Managed SSL cert v3 (api · admin · aegisimaging.ai · www) + Cloud Run (API · Dashboard · Landing · 10 sidecars)					
API Backend (Go / Cloud Run) - Upload orchestration (sessions, chunked PUT) - Study / project / institution management - Routing rules engine → auto-pipeline dispatch - DICOMweb proxy (QIDO-RS + WADO-RS) - Export shares: token-auth, ZIP download - Email digest scheduler + SMTP relay - DIMSE retry control proxy - MCP tool backend + batch import CLI - distroless image — minimal CVE surface		Admin Dashboard + dwvReact / Cloud Run - Behind Identity-Aware Proxy (IAP) - Study browser: filter, search, paginate, bulk ops - DWV — ydwwefore/after defacing review 7-stage pipeline visualization per study - RBAC: admin (write) + viewer (read-only) - Routing rules, institutions, anon profiles - Protocol templates, API keys, webhooks - Audit log + CSV export, share management - Federation peers, project lifecycle		Cloud SQL (PostgreSQL 15) - Studies, projects, institutions, admin users - Routing rules, audit trail, export shares - Private IP · Secret Manager credentials - Point-in-time recovery (7-day retention)	
Cloud Storage (GCS) - dicom/raw/{uid}/ — tag-de-identified - dicom/clean/{uid}/ — defaced + processed - bids/{uid}/ — NIFTI / BIDS output - analytics/{uid}/ — segmentation + metrics - Shared volume (Go API + all 10 sidecars)					
Processing Sidecars (Python / Cloud Run) — 9 services dispatched async by Go API, runs hands-free via routing rules					
Defacing (Python) - Head MRI / PET / CT — facial feature removal - mri_reface / DeepDefacer / mri_deface backends - nibabel fallback for dev / testing only - Writes defaced DICOM to clean/ store - SSIM-based QA score per study		PHI Detection (Python) - Burned-in text OCR on DICOM pixel data - Gemini / Cloud Vision / Azure / Textract / Tesseract - Pixel redaction — detect + mask burned-in PHI - LLM-based text scrubbing (Gemini / OpenAI / Anthropic) - Per-project confidence threshold config		QC Automation (Python) - File integrity + required DICOM tag check - Slice consistency (rows / cols / pixel spacing) - SNR estimation (signal mean / corner noise) - Coverage completeness by body part - Missing slice gap detection	
Classification (Python) - Fills modality + body_part from DICOM headers - Heuristic: tags → SOP UID → series description - Gemini / Cloud Vision / Rekognition fallback - Re-evaluates routing rules after classify - Confidence threshold gating (≥ 0.5)		BIDS Conversion (Python) - DICOM → NIFTI via dcm2nix - BIDS-compliant directory structure - sub-{hash8}/anat func dwi perf pet/ - JSON sidecar metadata per series - Privacy: UID-hashed subject labels			
Protocol Check (Python) - Verifies TR / TE / flip / thickness vs template - Classic + Enhanced DICOM (multi-frame) - Per-project protocol templates with CRUD - numeric / exact / range / contains_all match - critical / warning / info severity levels		Analytics (Python) — 18 Backends - Brain: FreeSurfer · SynthSeg · BrainSuite · volBrain · Atlas ROI - Spine: TotalSpineSeg · SPINEPS - Whole-body: TotalSegmentator · nnU-Net · MedSAM2 · MONAI Label - Specialized: PETSurfer · QSM · BASIL · FSL · ANTs · SPM · ITK-SNAP - Longitudinal: TBM-SyN + FreeSurfer Long stream		SCT — Spinal Cord Toolbox (Python) - Spinal cord segmentation (sct_deepseg_sc) - Vertebral labeling (sct_label_vertebrae) - CSA per vertebral level (sct_process_segmentation) - Compression metrics: aMCC, aSCOR - DTI mapping: FA, MD, AD, RD (optional DWI)	
Synth MRI (Python) ★ - Synthetic brain MRI generation - CPU: Shepp-Logan phantom (numpy + pydicom) - GPU: MONAI BraTS LDM (INCLUDE_MONAI=true) - T1w contrast, Rician noise, face anatomy - Seeded + reproducible, 20 – 200 slices		Landing Page (nginx / Cloud Run) - aegisimaging.ai — public marketing site - React + Vite built, served by nginx - Interactive synthetic DICOM demo widget - Schedule Demo / Contact form - Market opportunity: \$45B imaging market - LB default backend — no IAP required - www + apex domains on SSL cert v3			
Security Layers - VPC private subnets + Cloud NAT - Cloud Armor DDoS / WAF - TLS 1.2+ on all endpoints - CMEK (Cloud KMS) - IAM least privilege - Secret Manager (DB creds, API keys) - Cloud Audit Logs - Artifact Registry vuln scanning - distroless containers (Go API) - IAP on all admin routes - No PHI in email / audit entries		Export & Sharing Export Portal (React / public SPA) - Token-authenticated share links - Study info, expiry countdown (server-anchored) - ZIP download of approved DICOM files DICOM Forwarding (route_to) - DICOMweb STOW-RS destinations - DIMSE C-STORE to remote AE Title - Auto-forwards on study approval - Cross-cloud: GCP ↔ AWS STOW-RS live Share Lifecycle - Extend / revoke anytime - Immutable download log per share - Export analytics dashboard		MCP Server + AI Agent Tools Model Context Protocol (Claude / Cursor) 96 read tools: studies, audit, stats, routing, health 95 write tools (confirm:true + reason required) - Zod-validated schemas — read-safe by default - DIMSE retry proxy: process, replay, dead-letter Agent Orchestrator - DICOM tag provenance analysis - Diagnostic tools for study triage - Cohort report, pipeline funnel, project health Batch Import + Webhooks + API Keys - aegis-import CLI — bulk historical migration 5 webhook events, HMAC-SHA256 signed payloads - Machine-to-machine bearer tokens (hashed)	
AWS ACCOUNT <AWS_ACCOUNT_ID> · us-east-1 · ✓ Live — aws.api.aegisimaging.ai · Cross-cloud routing verified · Feb 25, 2026					
ALB + Cognito (Auth Layer) - HTTPS listener on ACM certificate - Cognito hosted UI — admin-create-only user pool - authenticate-cognito default action - Public bypass rules: /healthz, upload, export		API + Admin (ECS Fargate) - Go API — same image as GCP (1 vCPU / 2 GB) - Admin Dashboard — React / nginx (0.5 vCPU / 1 GB) - Service discovery: api.aegis.local - Force-new-deployment via GitHub Actions		9 Python Sidecars (ECS Fargate) - defacing · phi-detection · qc-service - bids-service · classification-service - protocol-service · synth-service - analytics-service · sct-service - Cloud Map private DNS: svc.aegis.local:8080	
RDS + S3 (Data Layer) - RDS PostgreSQL 15 — private subnet - Secrets Manager — master credentials - S3 DICOM bucket — versioned, KMS-encrypted - Same STORAGE_MODE=s3 as GCP (same Go code)		DIMSE EC2 (t3.small) - Elastic IP — stable for PACS AE title registration - Amazon Linux 2023 — Docker + SSM agent - SSM Parameter Store → image URI on every boot - GitHub Actions: write SSM param + reboot instance		GitHub Actions CI/CD - Triggers on push to develop — parallel with GCP Cloud Build - Matrix build: 11 services, --platform linux/amd64 - Push SHA tag + latest tag to 13 ECR repositories - Force-new-deployment for 10 ECS services - Update SSM param + reboot DIMSE EC2	
AZURE SUBSCRIPTION Container Apps · PostgreSQL Flexible Server · Azure Blob Storage · ✓ Live — Feb 26, 2026 · GitHub Actions OIDC CI/CD					
Azure Container Registry + Easy Auth - ACR — private registry, GitHub Actions OIDC push - Easy Auth on Container Apps — injects X-MS-CLIENT-PRINCIPAL-NAME - AUTH_PROVIDER=azure, AUTH_ENABLED=true - Federated OIDC — no long-lived credentials stored		API + Admin (Azure Container Apps) - Go API — same image as GCP/AWS (0.5–1 vCPU / 2 GB) - Admin Dashboard — React / nginx Container App - Azure Communication Services SMTP relay (smtp.azurecomm.net) - Force-new-revision via GitHub Actions on every push		9 Python Sidecars (Container Apps) - defacing · phi-detection · qc-service - bids-service · classification-service - protocol-service · synth-service - analytics-service · sct-service - Internal ingress only — same Docker images as GCP/AWS	
PostgreSQL Flex + Azure Blob (Data Layer) - Azure Database for PostgreSQL — Flexible Server - Azure Blob Storage — STORAGE_MODE=azure - DefaultAzureCredential — Workload Identity / Managed Identity		DIMSE Azure VM (Standard_B2s) - Debian 12 — Docker + Azure VM Extensions - Static public IP — TCP port 11112 for PACS registration - GitHub Actions: az vm run-command + docker pull/restart		GitHub Actions CI/CD (OIDC) - Triggers on push to develop — parallel with GCP + AWS - Federated OIDC — AZURE_CLIENT_ID / TENANT_ID / SUBSCRIPTION_ID - Build 13 images → push to ACR → az containerapp update	

Automated Processing Pipeline

routing rules define which steps are required → pipeline dispatches each step in dependency order, hands-free

1. Classify

fills modality
body_part

2. PHI Scan

OCR pixels
redact burned-in

3. Protocol

TR / TE / flip
vs template

4. Deface

remove face
head imaging

5. QC Check

SNR · coverage
slice consistency

6. BIDS

NIFTI + sidecar
BIDS structure

7. Analytics

18 backends
brain · spine · body

8. SCT

cord CSA
compression

9. Export Fwd

STOW-RS or
DIMSE forward

Implementation Phases

M1: Foundation + GCP ✓

Go API · 9 Python sidecars · Upload Portal · Admin Dashboard · PostgreSQL · Terraform · CI/CD

M2: AWS Live ✓

ECS Fargate · RDS · S3 · ALB + Cognito · Cross-cloud DICOM routing

M3: Azure Live ✓

Container Apps · PostgreSQL Flex · Azure Blob · ACR · GitHub Actions OIDC CI/CD

M4: Processing & Security ✓

JPEG2000 / Enhanced DICOM · Pixel redaction · Clinical trial RBAC · Security hardening

M5: Analytics & MIDI-B ✓

18 analysis tools · SCT · Longitudinal · Biomarker DB · MIDI-B compliance · Niivue

M6: Private Beta •

Invite-gated access · Feedback · DIMSE C-MOVE / C-FIND · SLA hardening

M7: Compliance ◦

BAA template · SOC 2 Type I audit · DIMSE at scale · AWS Marketplace

M8: Enterprise Integrations ◦

SOC 2 Type II · HL7 FHIR · Federation · DICOM conformance v2

M9: Enterprise GA ◦

On-premises · Multi-tenant SaaS · PACS/VNA native integration

Key Open-Source Dependencies

Go:

suyashkumar/dicom · pgx · testcontainers-go · testify

Browser:

dcmjs · dicomParser · dwvIT) · React 19 · Vite

Defacing:

mri_reface · DeepDefacer · mri_deface · dcm2nix · nibabel

PHI/OCR:

pytesseract · Google Cloud Vision · AWS Textract · Pillow

QC/BIDS:

pydicom · numpy · dcm2nix · pynetdicom (C-STORE SCP)

Analytics:

FreeSurfer · FSL · ANTs · SPM · SynthSeg · nnU-Net · TotalSegmentator · MONAI Label

SCT/Spine:

Spinal Cord Toolbox · TotalSpineSeg · SPINEPS · MedSAM2

Synth:

numpy · pydicom · MONAI Generative · torch · huggingface-hub

Infra:

Terraform Google + AWS + Azure providers · Docker distroless/slim

Multi-Cloud + Test Coverage

GCP:

Cloud Run · Cloud SQL · GCS · IAP · Cloud Armor · KMS · Pub/Sub

AWS:

ECS Fargate · RDS · S3 · ALB + Cognito · ACM · Secrets Manager

Local:

Docker Compose · PostgreSQL 15 · Mailpit · local filesystem

Auth:

GCP IAP · Azure AD Easy Auth · AWS ALB+Cognito · dev auto-auth

Storage:

STORAGE_MODE=gcs | s3 | azure | local — same Go API, no code changes

Azure:

Container Apps · PostgreSQL Flex · Azure Blob · ACR · Easy Auth · Azure VM (DIMSE)

CI/CD:

GitHub Actions: Go (1,072) · Python (892) · Client (166) · TS · Docker (15) · Cloud Build auto-deploy

Domains:

aegisimaging.ai · www · api · admin — SSL cert v3